

User Experience Engineering - Spring 2026

Milestone 1 Detailed Description

Initial Concept and Needs Finding Plans

In milestone 1, you will detail your project proposal by specifying your target users, the need you are addressing, and how you see the system you design improving things for your users.

You will also create a plan for your first UX research study in this project--a needs finding study. Turning in your plans before getting started will allow us to check in with you and make sure you're starting out on a good foot.

Summary of Activities and Outputs

Step	Activities	Output	Time
Clarify the design problem and scope	• Select design challenge • Problem research • Identify existing offerings • Web searches: background research • Informal interviews	(See project brief)	3 hours
Create project brief	• Define target audience • Problem definition • User needs • Activities • Overview of competitive landscape (preliminary) • System goals	Project brief	1 hour
Plan needs finding study	• Establish study goals • Develop recruiting criteria • Design interview protocol	Study plan	2 hours

Activity Descriptions

Activity 1: Clarify the design problem and scope

Step	Activities	Output	Time
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Clarify the design problem and scope	• Select design challenge • Problem research • Identify existing offerings • Web searches: background research • Informal interviews	(See project brief)	3 hours
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The first step will be to choose the problem you're going to work on. You should review the design challenges we've provided and decide which one to pursue, based on your interest and on whether you think you'll be able to access members of the target user population. The design challenges on their own, however, are not nearly specific enough to guide a project. You will have to do some work (mainly thinking, reflection, and some lightweight research including searching for competitors and informal "interviews" with potential users) to more clearly nail down a specific user population and a problem that you want to solve for that population.

Some things to think about include:

- Do you have access to members of the target population?
- Do current products and practices adequately fail to address the problem/need?
- Have you articulated the target population and problem/need clearly enough that you can imagine what it would look like for a design to be "successful" in addressing or solving the problem/need?

There is no concrete output from this activity, per se, but the results of your work here will form the core of your Project Brief.

Activity 2: Create a Project Brief

Step	Activities	Output	Time
Create project brief	• Define target audience • Problem definition • User needs • Activities • Overview of competitive landscape (preliminary) • System goals	Project brief	1 hour

Once you have settled on your the design problem, which includes surveying the competitive landscape, identifying a user population, and identifying a problem to be solved and/or a need to support. To create the brief, you just need to write this all down in a way that is clear enough to communicate to others why your design is needed and what goals it needs to satisfy. See details of the Project Brief in the description of Milestone Deliverables below.

Activity 3: Plan Needs Finding Study

Step	Activities	Output	Time
Plan needs finding study	- Establish study goals - Develop recruiting criteria - Design interview protocol	Study plan	2 hours

Now that you've created your project brief, it's time to get to work! The first step will be to conduct a study to better understand, in detail, your users' needs with respect to the activities that your system will support.

For this activity, you will plan your study, including the following:

- Establishing your high-level study goals:** these are the questions you want to answer with your study. Most likely, the key question will be
 - "How do <users in my population> currently do <activity>?" along with
 - "What problems do <users in my population> face when trying to accomplish <activity>?" and
 - "What goals, values, other activities, relationships, and constraints (e.g., applications and platforms currently in use, money, time, technical expertise) inform how and
 - Why <users in my population> do <activity> the way they do?"

Depending on particulars, you may have additional, more specific questions to add as well.

- Defining your recruiting criteria and plans:** Nail down your specific criteria for recruiting participants in your needs finding study. Your recruitment criteria ought to look into Expertise, Characteristics, Behaviors and Attitudes of your target users. You should plan to interact with *at least 4 potential users* of your proposed system. The interactions should include an interview of at least 30 minutes, and may include observations as well. If possible, you should identify particular individuals you plan to ask to participate, though you don't need to contact them.
- Designing the interview protocol:** Your needs finding study must contain interviews with at least 4 people. For the interview protocol, you should follow the guidelines laid out in Understanding User Needs lesson, including the use of a semi-structured approach, the incorporation of observations along with direct questions (in this case you'll be observing how they do things now, using whatever tools or methods they happen to use), and good questions vs not-so-good questions. Reviewing these guidelines is *strongly recommended* before putting together your protocol. If you deviate from the recommended protocol for some reason (e.g., not including observations) you should explain this decision in your protocol.

Milestone Deliverables

Produce a single document (.pdf) with two clearly marked sections: Design Brief, and Needs Finding Study Plan.

Section 1: Design Brief

The Design Brief section should include:

- A description of the target user population

- A description of the problem to be solved, including the high-level user/community activities that need to be supported (e.g., “grocery shopping” or “garden tool sharing”)
- A brief, preliminary overview of the “competitive landscape,” i.e., the tools and methods that already exist for supporting the target activity (which can include both digital tools like apps and “analogue” methods like face-to-face contact or paper lists)
- A statement of system goals, including what the system will allow users to do (a preliminary list of features/capabilities) and why you believe it will be better than the current way of doing things.

Section 2: Needs Finding Study Plan

The Needs Finding Study Plan should include:

- The high-level goals of the study (1-2 sentences, with key questions)
- A description of the recruiting criteria and strategy.
- The interview protocol (following the template provided as part of the Interview Protocol in Understanding User Needs Lesson)

You will upload this file on the “Project Milestone 1 Submission” page on Google Classroom.

Submission Instructions

Please submit a pdf file with the name format:

Spring-2026-M01-YourTeamName.pdf

The pdf must be submitted on google classroom.

Make sure your Team No, Team Name, member names, roll numbers and the milestone title is clearly mentioned on the title page of the deliverable that you submit.

If your submission is late or doesn't follow the naming scheme/instructions, it will not be graded.